

KARTHIK VISWANATHAN

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EDUCATION	University of Amsterdam (UvA) 2021 - 2025 PhD in Physics Specialization: Application of Topological Data Analysis to Physical Systems Advisor: Dr. Jan Pieter van der Schaar Expected Graduation: 31 December 2025
	University of Amsterdam (UvA) 2019 - 2021 MSc in Physics and Astronomy, track Theoretical Physics, cum laude GPA: 8.7/10
	Indian Institute of Technology Madras (IITM) 2013 - 2017 B.Tech in Engineering Physics GPA: 8.29/10
RESEARCH	Masters Thesis 2020 - 2021 Exploring the Spectral Theory/Topological Strings duality Advisor: Dr. Marcel Vonk
EXPERIENCE	Bachelors Thesis 2017 Real Space Renormalization and Applications to Machine Learning Advisor: Dr. Ashwin Joy
PROFESSIONAL EXPERIENCE	Surveillance Analyst at Goldman Sachs, Bangalore 2017 - 2019 Developed ML models for anomaly detection in financial data Manager: Prof. Howard Karloff
	Summer Internship at Goldman Sachs, Bangalore 2016 Fast Personalized PageRank in a MapReduce framework Mentor: Dr. Koushik Balasubramanian
PUBLICATIONS	K. Viswanathan , Y. Gardinazzi, G.Panerai, A. Cazzaniga and M. Biagetti The Geometry of Tokens in Internal Representations of Large Language Models arXiv:2501.10573 [cs.CL] Y.Gardinazzi*, G.Panerai*, K. Viswanathan* , A.Ansuini, A.Cazzaniga and M.Biagetti Persistent Topological Features in Large Language Models arXiv:2410.11042 [cs.CL] J.H.T. Yip, M. Biagetti, A. Cole, K. Viswanathan and G. Shiu Cosmology with persistent homology: A Fisher forecast JCAP 09 (2024), 034, arXiv:2403.13985 [astro-ph.CO]
AWARDS AND DISTINCTIONS	Sander Bais Prize to the Academic Merit 2020 Awarded by Institute for Theoretical Physics Amsterdam for exceptional academic performance in the master's program.
	ACM ICPC World Finals May 20, 2015 Represented India in the international collegiate programming contest. Marrakech, Morocco
	International Olympiad in Informatics Training Camp May, 2013 Selected for the Informatics Camp after ranking in the top 25 in the Indian National Olympiad in Informatics. Bangalore, India

* - These authors contributed equally to this work

TECHNICAL SKILLS	Programming Skills: PyTorch, TensorFlow, Hadoop, MapReduce, Mathematica	
	Code Repositories:	
	1. https://github.com/RitAreaSciencePark/token_geometry 2. https://github.com/RitAreaSciencePark/ZigZagLLMs	
	Computational Resources: I have acquired expertise in analyzing cosmological N-body simulations and internal representations in transformers using geometric and TDA-based methods. To analyze these datasets, I have successfully applied for GPU and CPU time both at the LUMI consortium and at the National Dutch Computing Facilities (Snellius supercluster).	
ORGANIZATION AND MENTORING	Organizer of the Cosmology Journal Club	2022
	University of Amsterdam	
	Mentoring of Master Student	2023
	Master's Degree Candidate: Sibilla Bouche	University of Amsterdam
	Thesis Title: The persistence of non-Gaussian features: a Neural Ratio Estimation approach	
	Supporting Master Student	2023
	Master's Degree Candidate: Sliem el Ela	University of Amsterdam
	Thesis Title: From Primordials to Persistence: A Dual Exploration of Multiparameter Topology and Cosmic Origins	
	Supporting Master Student	2024
	Master's Degree Candidate: Giada Panerai	AREA Science Park, Trieste
	Thesis Title: Zig-Zag Persistence in Neural Networks Representations	
TEACHING	Teaching Assistant	2020 - 2024
	MSc Physics Program, University of Amsterdam	
	Courses: Topological Data Analysis: a Physics Perspective, Advanced Cosmology: Non-linear Structure Formation and Observations, ML for Physics and Astronomy, QFT in Curved Spacetimes, General Relativity	
RESEARCH VISITS	Area Science Park, Trieste, Italy	September 16 - October 4, 2024
	Area Science Park, Trieste, Italy	February 15 - April 27, 2024
CONFERENCES AND WORKSHOPS	Short Talk	December 6, 2024
	PhD and PostDoc Symposium	Amsterdam, Netherlands
	Title: Geometry of Internal Representations in Large Language Models	
	Short Talk	July 1 - July 5, 2024
	New Strategies for Extracting Cosmology from Galaxy Surveys	Sexten, Italy
	Title: Information Maximizing Persistent Homology for Halo Catalogs	
	Long Talk	June 5 - June 9, 2023
	Applications of Topological Data Analysis to Cosmology and Beyond	Trieste, Italy
	Title: Information Maximizing Persistent Homology	
	Long Talk	June 27 - July 1, 2022
	Interpretable and Higher-Order Statistics for Late-Time Cosmology	Trieste, Italy
	Title: Antifragile Persistent Homology using Fisher Information	

Poster	May 16 - May 17, 2024
Trends in Theory Symposium	Wageningen, Netherlands
Title: The Geometry and Physics of Hidden Representations in LLMs	
Poster	June 26 - June 30, 2023
Danish-Swedish Summer School on TDA and Spatial Statistics	Aalborg, Denmark
Title: Information Maximizing Persistent Homology for Inference	
Participant	June 6, 2024
Complexity Across Scales	Amsterdam, Netherlands
Participant	May 13–May 17, 2024
Dutch Research School of Theoretical Physics	Wageningen, Netherlands
Participant	September 12–September 16, 2022
Physics meets Artificial Intelligence	Munich, Germany
Participant	July 18–July 22, 2022
Young Topologists Meeting	Copenhagen, Denmark
Participant	October 4–December 17, 2021
Amsterdam-Brussels-Geneva-Paris Doctoral School	

REFERENCES

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